

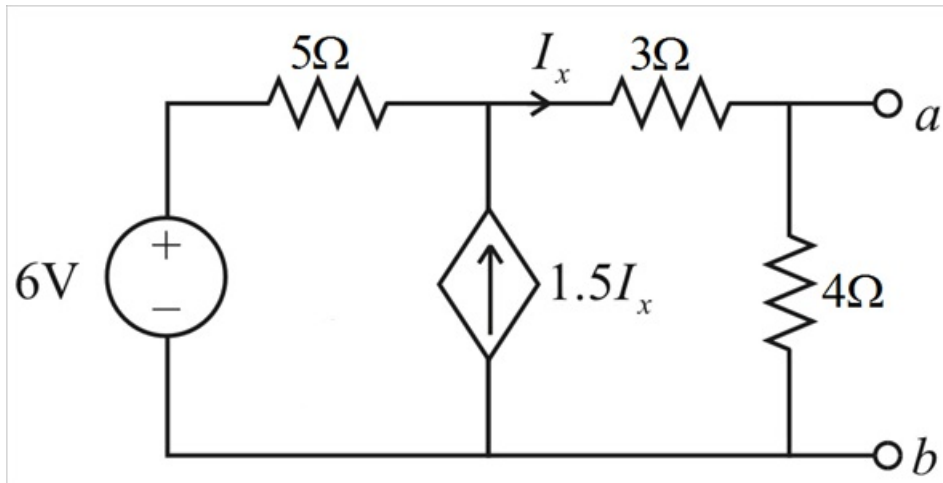
Problem

Rework Practice Prob. 4.9 using *PSpice*.

Step-by-step solution

Step 1 of 6

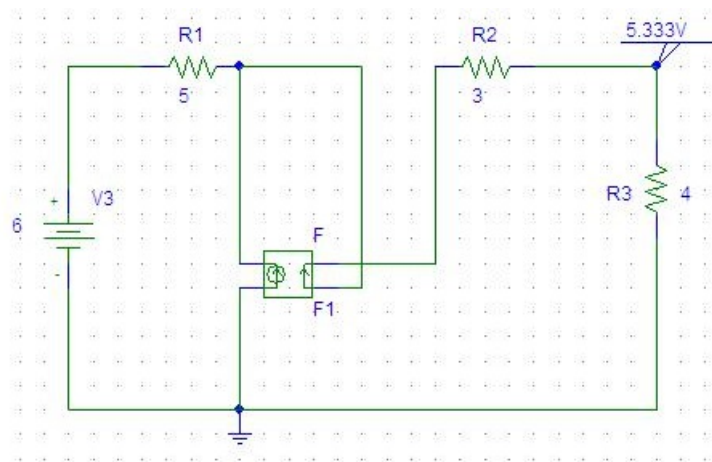
Considering the circuit



Step 2 of 6

V_{th} can be found by finding voltage between the terminals 'ab'

Now draw the circuit using schematics



Step 3 of 6

To find the voltage between terminal 'ab' pseudo component view point are added to display the required node voltage.

$$\therefore V_{ab} = 5.333\text{V}$$

$$V_{ab} = V_{th}$$

$$\therefore V_{th} = 5.333\text{V}$$

Step 4 of 6

To find the thevenin resistance R_{th} $a - b$ in the circuit.

Place the test current source across the terminals 'ab'

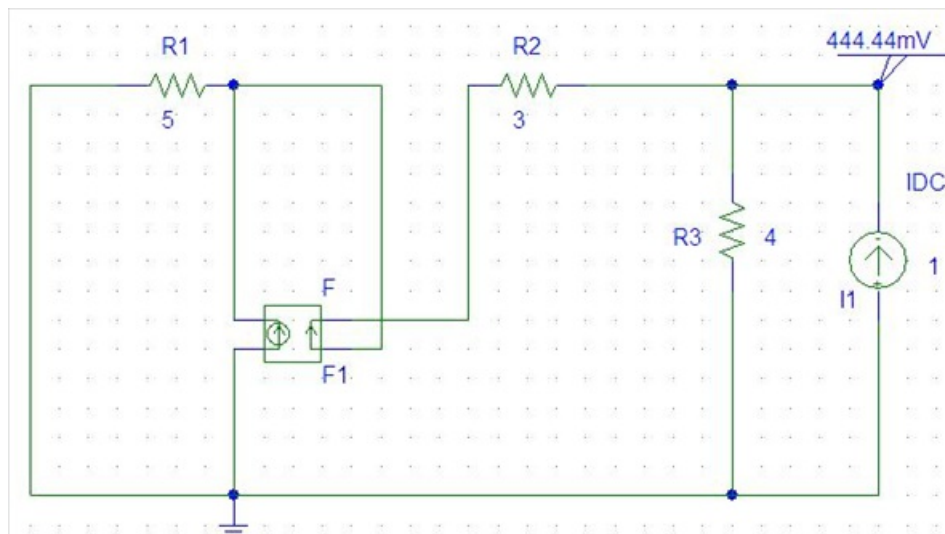
R_{th} can be found by finding the voltage across the test current source

$$\therefore R_{th} = \frac{V_{ab}}{1\text{A}}$$

$$R_{th} = V_{ab}$$

The pseudo component view point is added to display the required node voltage and Simulated circuit diagram is

Step 5 of 6



Step 6 of 6

$$\therefore R_{th} = \frac{0.444}{1}$$

$$R_{th} = 0.44\Omega$$